MarketSage AI

Master Engineering Blueprint — **Part VII**

Repository, Resources, Risk and Appendices

Monorepo layout, engineering conventions, team shapes, cost structure, the risk register, reference formulas, the taxonomy skeleton and glossary.

This is an extract of the MarketSage AI master plan, published separately for convenience. Chapter, section and equation numbering matches the full document, and cross-references point back into it. Read Chapter 3 (Reality Calibration) first — every design decision in this part follows from it.

CHAPTER 18

Repository Structure and Engineering Conventions

Listing 18.1: Monorepo layout

```
marketsage/
  docs/
    marketsage_master_plan.tex # this document
    gates.md # SIGNED phase-gate records with numbers
    decisions/ADR-0001-broker.md # architecture decision records
    compliance_memo.md
  config/
    instruments.yaml costs.yaml # dated, point-in-time cost tables
    risk_limits.yaml # changes require an ADR
    sources.yaml # source registry + credibility weights
    taxonomy.yaml # versioned event taxonomy
  libs/
    ms_core/ # schemas, clock, bus client, tracing, config loader
    ms_data/ # tick handler, candle builder, chain snapshotter, QC
    ms_news/ # collectors, normaliser, dedup, publish-time resolver
    ms_events/ # classifier, severity, novelty, confirmation, KG routing
    ms_impact/ # event study, abnormal returns, Bayesian posteriors
    ms_features/ # feature registry, offline+online materialisers, linter
    ms_models/ # training, CV harness, calibration, conformal, registry
    ms_options/ # IV surface, greeks, structures, Monte Carlo, sizing
    ms_risk/ # limits, kill switches, portfolio state
    ms_sim/ # L2 event-driven simulator, fill + slippage models
    ms_broker/ # broker adapters (paper + live), order state machine
  services/
    ingest_market/ ingest_news/ engine_event/ engine_predict/
    engine_decide/ api_gateway/ ui_web/
  jobs/
    nightly_features.py weekly_retrain.py event_study_rebuild.py
    postsession_outcomes.py morning_briefing.py
  tests/
    unit/ replay/golden_day_2026_07_14/ stress/ parity/
  notebooks/ # research only; never imported by services
```

Convention	Rule
No magic numbers	Every threshold lives in `config/`, is loaded at startup, and its hash is logged with each decision.
Schemas first	Every bus message has a versioned schema; consumers validate. Adding a field is allowed, changing a meaning requires a version bump.
Notebooks are disposable	A notebook may never be imported by a service. If code matters, it moves into a library with tests.
Time is explicit	No implicit `now()` in library code — the clock is injected, so the simulator and production share the same code path.
Live requires intent	Any function that can place a real order takes `live: bool = False` and logs loudly.
Tests that matter	Determinism test, offline–online parity test, leakage linter, golden-day replay, stress scenarios. These run in CI on every commit.
Gate discipline	`docs/gates.md` is the project's spine. No gate entry, no phase advance.

CHAPTER 19

Resources, Cost and Risk Register

19.1 Team shapes

Shape	Timeline to G8	Notes
Solo (you)	45–60 weeks	Entirely feasible. The binding constraint is data-engineering grind, not modelling. Expect Phases 1–2 to feel slow and unglamorous; they are the moat.
Two people	30–38 weeks	Optimal split: one owns data and infrastructure (P0–P2, P7 simulator, P8 ops), the other owns ML and research (P3–P6).
Three to four	24–30 weeks	Add a dedicated frontend and a research generalist. Beyond four, coordination cost exceeds benefit at this stage.

19.2 Cost structure (relative, not absolute)

Prices change; the *ratios* do not. Plan the budget in these proportions.

Item	Share of year-1 spend	Comment
Historical option chain data	**Largest single item**	Unavoidable for a credible options backtest. Buy once, clean carefully, treat as an asset.
Live market data (broker or vendor)	Moderate, recurring	Start on the broker feed; upgrade only when the gate demands it.
News and social API tiers	Moderate, recurring	Start with 3–5 high-value sources rather than breadth.
Compute (cloud VM in the correct region, storage)	Moderate, recurring	One solid VM plus fast disk beats a Kubernetes cluster at this stage.
GPU for training	Small	Rent by the hour for training runs; do not buy or keep one running. Inference runs on CPU via ONNX.
Trading capital at R1	Deliberately tiny	Capital is the *last* thing you spend, not the first.
Legal/compliance consultation	Small, one-off	Cheaper than one regulatory problem.

Budget principle

Spend on **data** first, **time** second, **compute** third and **trading capital** last. Most failed projects invert this exactly.

19.3 Risk register

Risk	Prob	Impact	Mitigation
No intraday directional edge exists at our data tier	High	High	Gate G5 tests this at week 24 for a fraction of the total cost. Pivot options are pre-defined: volatility-only, or the gap product.
Historical option chain data unavailable or unaffordable	Med	High	Identify vendors and prices in Phase 0, not Phase 5. If truly blocked, restrict v1 to futures and the gap product.
Backtest overfitting	**Very high**	**Very high**	Purged CV, embargo, combinatorial CV, deflated Sharpe, locked holdout evaluated once, experiment log. This is the default outcome unless actively fought.
Live results diverge from paper	High	High	Run paper alongside live permanently; the gap is measured, not guessed. Capital ladder demotes instantly.
Slippage larger than modelled	High	Med	Calibrate the slippage model from your own fills; require expected value to clear costs with margin.
Fake or retracted news triggering a trade	Med	**Very high**	Multi-source confirmation, severity gating, retraction handling, the AP-hack CI test.
Regime change invalidating models	High	Med	Regime-conditional models, drift monitors, automatic class disabling, scheduled retrains.
Broker API outage or rejection mid-position	Med	High	Order state machine with reconciliation, flatten-on-failure drills, a manual override runbook.
Regulatory change affecting retail algo access	Med	High	Compliance memo maintained; architecture keeps the human-approval mode as a first-class path, not a hack.
Data licensing violation when adding users	Med	High	Legal review before Tier T5. Do not show live exchange prices to third parties without a licence.
Solo-developer burnout	**High**	High	Phase 4 ships a real product at week 18. Weekly gates give visible progress. The plan is explicitly designed so that stopping at any gate still leaves something valuable.
Scope creep (adding stocks, crypto, more countries)	Very high	Med	Chapter 2 scope decisions are frozen until G8. Write new ideas in a backlog file; do not build them.

19.4 What "done" looks like at each tier, restated as artefacts

Tier	The artefact that proves it
T0	A year of complete, replayable, point-in-time data on disk with automated quality reports.
T1	An automated daily note explaining yesterday's moves, rated by you against reality for 20 sessions.
T2	A reliability diagram from live paper trading with a slope near 1 and a positive Brier skill score.

T3	A signed G8 record: 60 sessions, 150 trades, positive expectancy after costs, live matching simulated.
T4	A broker statement showing real fills on ladder rung R1, and a paper-versus-live divergence chart within tolerance.
T5	A legal opinion, a data licence, a multi-tenant deployment and an SLO report.

APPENDIX A

Reference Formulas

Black–Scholes / Black-76 Greeks

With $d_1 = \dfrac{\ln(F/K) + \frac{1}{2}\sigma^2 T}{\sigma\sqrt{T}}$, $d_2 = d_1 - \sigma\sqrt{T}$, discount factor $D = e^{-rT}$:

$$C = D[F\,N(d_1) - K\,N(d_2)], \qquad P = D[K\,N(-d_2) - F\,N(-d_1)],$$

$$\Delta_C = D\,N(d_1), \qquad \Gamma = \frac{D\,\phi(d_1)}{F\sigma\sqrt{T}},$$

$$\mathcal{V} = D\,F\,\phi(d_1)\sqrt{T}, \qquad \Theta_C = -\frac{DF\phi(d_1)\sigma}{2\sqrt{T}} - rDKN(d_2) + rDFN(d_1).$$

Intraday scaling of implied volatility

$$\sigma_{h\text{ minutes}} = \sigma_{\text{annual}}\sqrt{\frac{h}{375 \times 252}}, \qquad \text{Implied move}_h = S\,\sigma_h.$$

Range-based realised volatility (Garman–Klass)

$$\sigma^2_{GK} = \tfrac{1}{2}\big(\ln\tfrac{H}{L}\big)^2 - (2\ln 2 - 1)\big(\ln\tfrac{C}{O}\big)^2.$$

HAR-RV volatility model

$$RV_{t+1} = \beta_0 + \beta_d RV_t^{(d)} + \beta_w RV_t^{(w)} + \beta_m RV_t^{(m)} + \varepsilon_{t+1},$$

with daily, weekly and monthly averages of realised variance. Add a gradient-boosted model on the residual with event features.

Order-flow imbalance

$$OFI_t = \sum_{i \in \text{window}} \big[\mathbb{1}_{\{P_i^b \geq P_{i-1}^b\}} q_i^b - \mathbb{1}_{\{P_i^b \leq P_{i-1}^b\}} q_{i-1}^b - \mathbb{1}_{\{P_i^a \leq P_{i-1}^a\}} q_i^a + \mathbb{1}_{\{P_i^a \geq P_{i-1}^a\}} q_{i-1}^a\big].$$

Pinball (quantile) loss

$$L_\alpha(y, \hat{q}_\alpha) = \begin{cases} \alpha\,(y - \hat{q}_\alpha) & y \geq \hat{q}_\alpha \\ (1-\alpha)(\hat{q}_\alpha - y) & y < \hat{q}_\alpha \end{cases}$$

Continuous ranked probability score

$$\text{CRPS}(F, y) = \int_{-\infty}^{\infty} \big(F(x) - \mathbb{1}_{\{x \geq y\}}\big)^2 dx \;\approx\; \frac{1}{N}\sum_i |x_i - y| - \frac{1}{2N^2}\sum_{i,j} |x_i - x_j|$$

for N sampled paths x_i — directly computable from the Monte Carlo output.

Brier score decomposition

$$BS = \underbrace{\text{Reliability}}_{\text{lower is better}} - \underbrace{\text{Resolution}}_{\text{higher is better}} + \underbrace{\text{Uncertainty}}_{\text{fixed by the data}}$$

Fractional Kelly

$$f^* = \frac{pb - (1-p)}{b}, \qquad f_{\text{used}} = \min(0.25 f^*, f_{\max}).$$

Deflated Sharpe ratio (concept)

Given N independent strategy trials, the expected maximum Sharpe under the null grows roughly with $\sqrt{2 \ln N}$. Always report how many variants were tested; an unreported trial count makes any Sharpe ratio meaningless.

APPENDIX B

Event Taxonomy Skeleton (YAML)

Listing B.1: config/taxonomy.yaml — abridged

```yaml
version: 1
families:
  monetary:
    classes:
      rate_decision: {sev_range: [6,10], scheduled: true, horizons: [5,15,30,60]}
      stance_change: {sev_range: [5,9], scheduled: true}
      liquidity_measure: {sev_range: [4,8], scheduled: false}
      unscheduled_action: {sev_range: [8,10], scheduled: false}
  macro:
    classes:
      inflation_release: {sev_range: [4,8], scheduled: true, needs_consensus: true}
      growth_release: {sev_range: [4,8], scheduled: true, needs_consensus: true}
      trade_balance: {sev_range: [3,6], scheduled: true, needs_consensus: true}
  geopolitical:
    classes:
      military_action: {sev_range: [6,10], scheduled: false, require_sources: 2}
      escalation_rhetoric:{sev_range: [3,7], scheduled: false, require_sources: 2}
      sanctions: {sev_range: [4,8], scheduled: false}
  energy:
    classes:
      production_quota: {sev_range: [5,9], scheduled: true}
      supply_disruption: {sev_range: [4,9], scheduled: false}
  flows:
    classes:
      expiry_pinning: {sev_range: [3,6], scheduled: true, intraday_only: true}
      oi_shift: {sev_range: [3,6], scheduled: false}
  noise:
    classes:
      opinion: {sev_range: [0,2], tradeable: false}
      repetition: {sev_range: [0,1], tradeable: false}

gates: # applied before any directional signal
  min_severity: 5.0
  min_credibility: 0.75
  min_novelty: 0.45
  max_already_priced: 0.6
  min_effective_sample: 8
```

APPENDIX C

Glossary

Term	Meaning in this document
Abnormal return	Return minus what was expected given time of day, regime and day type. The correct unit for event studies.
Calibration	Whether stated probabilities match realised frequencies.
Conformal prediction	A method producing prediction intervals with distribution-free finite-sample coverage guarantees.
CRPS	A proper scoring rule for a full predictive distribution.
Dealer gamma exposure	Aggregate options-dealer gamma; when negative, hedging amplifies moves.
Edge realisation	The regression slope of realised edge on predicted edge. A slope near zero means no usable edge, regardless of hit rate.
Embargo	A buffer period dropped after each test fold to prevent leakage from serial correlation.
Implied move	The move the option market has already priced for a horizon.
IV crush	A fall in implied volatility that can make a directionally correct option trade lose money.
Meta-labelling	A second model that decides whether to act on the first model's signal.
Novelty	Whether an item carries genuinely new information not already priced.
Point-in-time	Data as it was actually knowable at a past moment, including timestamps and later revisions.
Purged CV	Cross-validation that removes training samples whose outcome windows overlap the test window.
Regime	A market state (volatility, trend, liquidity) that conditions all other models.
Triple barrier	Labelling by which of the upper, lower or time barrier is hit first.
Volatility risk premium	Implied volatility minus forecast realised volatility; the core edge for volatility structures.

APPENDIX D

Final Word

The three sentences that matter

1. **Start the recorder tonight.** Models can be rebuilt; history cannot be recovered.
2. **Gate G5 at week 24 is the real project.** Everything before it exists to make that test honest; everything after it exists to exploit a positive answer safely.
3. **Calibration, not accuracy, is the product.** A system that knows what it does not know is the only kind that survives contact with a live market.

This blueprint is deliberately harder than the original concept, because the original concept was designed to impress and this one is designed to survive. The vision is unchanged: watch the world, understand the impact, forecast the distribution, recommend the trade, and score yourself in public. The difference is that every claim here has a test attached to it, and every test has a date.